New Attempt

- Due Nov 13, 2024 by 2:59am
- Points 10
- Submitting a text entry box, a website url, or a file upload
- Robot Project

Robot Project Challenge 2

- **Overview:**

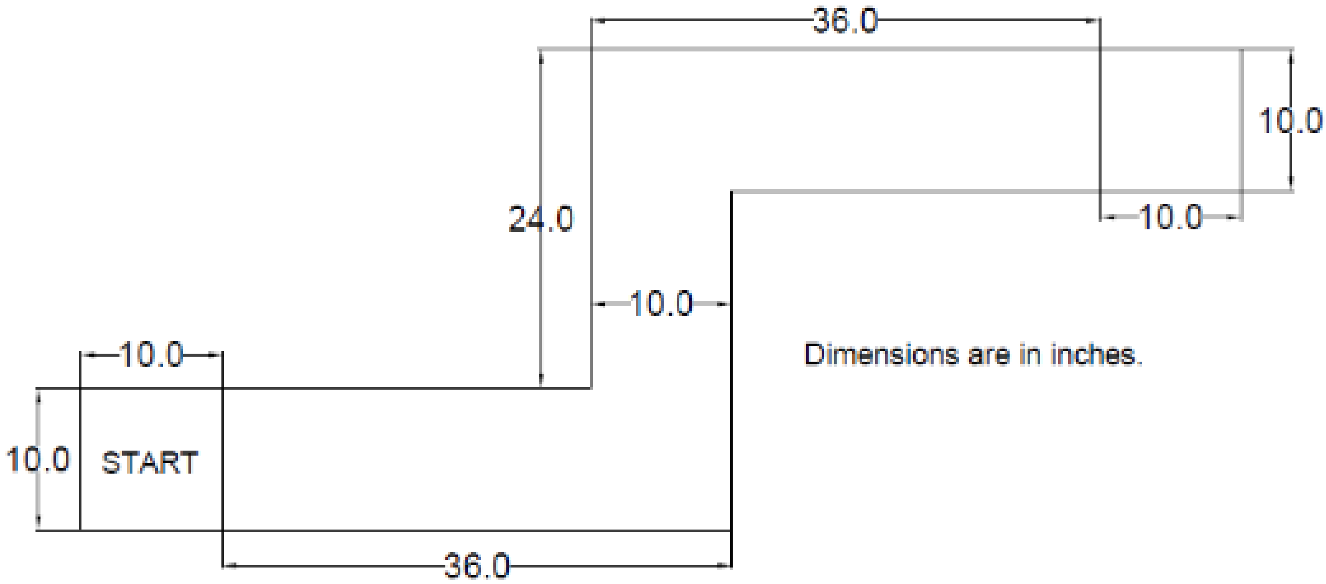
This is your second team challenge! Begin by meeting with your team to discuss the successes and challenges faced in Challenge 1. Together, program your robot to navigate a specified course, without using any inputs from the Serial Monitor. Be sure to record each step of the process and document all relevant details in your digital design notebook. Additionally, you will receive an individual peer-review assignment to evaluate another team's digital design notebook.

Reminder: It's crucial to dedicate time to your **Individual laser cutting project (https://northeastern.instructure.com/courses/189179/assignments/2352195)** and incorporate the best design of your team into your robot, ideally by **Challenge 3.**

- **Tasks:**
  - **Task 1: Reflection: As a team, reflect on the followings and record your insights in your design notebook:**
    - Review the successes and failures encountered during Challenge 1, and reflect on what you learned from each aspect.
    - Reflect on your teamwork experience during Challenge 1. Are there any concerns about the level of effort or time each member contributed? Are you satisfied with the quality of work produced by both your teammates and yourself? Based on these reflections, set specific goals to improve teamwork and team dynamics for the next challenge.
  - **Task 2**: **Program the robot to drive through the specified course shown below using components from the Sparkfun kit. You'll need to pre-program your robot, as there are no walls for any sensor to detect.**
    - Use button/switch/some other trigger to begin the program.
    - Operated automatically (you do not send commands from the computer).
    - Able to make right turn
    - Able to make left turn
    - Able to travel within the 10" boundary without going over the tape

- Stops at the end of the course
- Look at the code in Projects 5B and 5C and think about how you will program your robot to do this.
- I suggest recording all of your attempts on video and submitting the best one.

Robot Course:



- **Task 3: Record everything in your digital design notebook and share the link with Professor. Make sure the link is accessible and does not require permission. You can test this by copy and pasting the link to an incognito window.**
  - Your reflections from task 1
  - Flowchart your algorithm for programming the robot using draw.io and submit
  - Record a video of your best attempt (can be from during class or out of class)
    - You can include a link to the video on YouTube, Google drive, or other cloud service if you can't/don't want to upload the actual video
  - Plenty of pictures
  - Meeting minutes
  - Any success or failure
  - Be sure to update your digital design notebook every time you work on your robot project individually or as a team.
- **Task 4: Peer-review digital design notebook.** You will be assigned another team’s digital design notebook to review individually. Consider the followings for effective peer-reviewing:
  - **Read Thoroughly**
  - **Provide Constructive Feedback**: Use positive language and focus on specific elements. Comment on strengths (e.g., “Good clarity in explaining design changes”).Suggest improvements (e.g., “Consider adding more detail to your testing methods”).

- **Check for Completeness**: Make sure the notebook includes all required elements, as outlined in the assignment guidelines. If any section seems missing or incomplete, kindly note it in your feedback.
- **Be Specific and Clear**: Avoid vague feedback. Instead of "This part is confusing," say, "Consider rephrasing this explanation to clarify your design choices."
- **Use the Comment Box Feature**: Instead of editing the document directly, leave comments for each section where you have feedback or suggestions. Be concise but clear.
- **Summarize Key Feedback**: At the end and in the comment box section of peer-review assignment, include a brief summary of your feedback that highlights both strengths and main areas for improvement.

File Submission

As a group, submit the followings:

- Provide an accessible link to your digital design notebook.
- Arduino code
- Individually complete the digital design notebook assigned to you

Robot Challenge 2			
Criteria	**Ratings**		**Pts**
Robot Success (in-person or video) 1 pts each (5 requirements: button, left, right, drives straight, stops)	**5 pts** **Full Marks**	**0 pts** **No Marks**	5 pts
Code uploaded and compiles	**2 pts** **Full Marks**	**0 pts** **Does not compile or has errors**	2 pts
Digital Design Notebook Details Has all of the details requested	**2 pts** **Full Marks**	**0 pts** **No Marks**	2 pts
Peer review completed	**1 pts** **Full Marks**	**0 pts** **No Marks**	1 pts
			Total Points: 10